

Thomas Zimmer

其他

基本信息

姓名: Thomas Zimmer

所属院校: 波尔多大学

地区: France

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

射频集成电路

EDA工具与电路验证

纳米电子与先进工艺

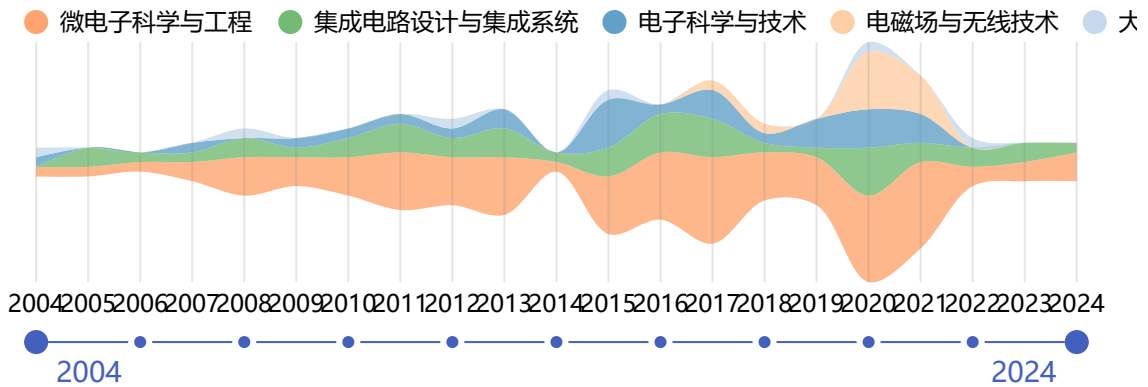
模拟与数模混合电路

模拟与射频IC设计

个人简介

Thomas Zimmer is a faculty member at the Laboratoire IMS, Université de Bordeaux. He has 13 followers, 2 followings, and 2 co - authors. His research interests are mainly in semiconductor device technology, especially Si/SiGe HBTs. He has published numerous papers, such as "28nm FD-SOI BiCMOSのための進歩したSi/SiGeH BTアーキテクチャ" in 2016, "Optimized Ring Oscillator With 1.65 - ps Gate Delay in a SiGe:C HBT Technology" in IEEE Electron Device Letters in 2013, and many others on topics like device modeling, calibration, and high - frequency measurement.

近年发表的论文



Exploring Compact Modeling of SiGe HBTs in Sub-THz Range With HICUM

IEEE TRANSACTIONS ON ELECTRON DEVICES

Panda, Soumya Ranjan; Zimmer, Thomas; Chakravorty, Anjan; Derrier, Nicolas; Fregonese, Sebastien

被引用次数: 0 2024

Establishing On-Wafer Calibration Standards for the 16-Term Error Model: Application to Silicon High-Frequency Transistor Characterization

IEEE JOURNAL OF MICROWAVES

Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 0 2024

Next Generation SiGe HBTs for Energy Efficient Microwave Power Amplification (Invited)

8TH IEEE ELECTRON DEVICES TECHNOLOGY & MANUFACTURING CONFERENCE, EDTM 2024

Panda, Soumya Ranjan; Billy, Philippine; Gauthier, Alexis; Guitard, Nicolas; Chevalier, Pascal; De Matos, Magali; Zimmer, Thomas; Fregonese, Sebastien

被引用次数: 0 2024

A TCAD-Based Analysis of Substrate Bias Effect on Asymmetric Lateral SiGe HBT for THz Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Panda, Soumya Ranjan; Fregonese, Sebastien; Chevalier, Pascal; Chakravorty, Anjan; Zimmer, Thomas

被引用次数: 2 2023

Electro-Thermal Investigation of SiGe HBTs: A Review

2023 IEEE BICMOS AND COMPOUND SEMICONDUCTOR INTEGRATED CIRCUITS AND TECHNOLOGY SYMPOSIUM, BCICTS

Zimmer, Thomas; Fregonese, Sebastien; Chakravorty, Anjan

被引用次数: 0 2023

A Technique for the In-Situ Experimental Extraction of the Thermal Impedance of Power Devices

IEEE TRANSACTIONS ON POWER ELECTRONICS

Scognamillo, Ciro; Fregonese, Sebastien; Zimmer, Thomas; d'Alessandro, Vincenzo; Catalano, Antonio Pio

被引用次数: 7 2022

BEOL Thermal Resistance Extraction in SiGe HBTs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Nidhin, K.; Balanethiram, Suresh; Nair, Deleep R. R.; D'Esposito, Rosario; Mohapatra, Nihar R. R.; Fregonese, Sebastien; Zimmer, Thomas; Chakravorty, Anjan

被引用次数: 1

2022

Optimizing Finger Spacing in Multifinger Bipolar Transistors for Minimal Electrothermal Coupling

IEEE TRANSACTIONS ON ELECTRON DEVICES

Gupta, Aakashdeep; Nidhin, K.; Balanethiram, Suresh; Yadav, Shon; Fregonese, Sebastien; Zimmer, Thomas; Chakravorty, Anjan

被引用次数: 0

2022

Performance prediction of InP/GaAsSb double heterojunction bipolar transistors for THz applications

JOURNAL OF APPLIED PHYSICS

Wen, Xin; Arabhavi, Akshay; Quan, Wei; Ostinelli, Olivier; Mukherjee, Chhandak; Deng, Marina; Fregonese, Sebastien; Zimmer, Thomas; Maneux, Cristell; Bolognesi, Colombo R.; Luisier, Mathieu

被引用次数: 4

2021

Reliable Technology Evaluation of SiGe HBTs and MOSFETs: fMAX Estimation From Measured Data

IEEE ELECTRON DEVICE LETTERS

Saha, Bishwadeep; Fregonese, Sebastien; Heinemann, Bernd; Scheer, Patrick; Chevalier, Pascal; Aufinger, Klaus; Chakravorty, Anjan; Zimmer, Thomas

被引用次数: 5

2021

Meander-Type Lines: An Innovative Design for On-Wafer TRL Calibration for mmW and Sub-mmW Frequencies Measurements

IEEE TRANSACTIONS ON TERAHERTZ SCIENCE AND TECHNOLOGY

Cabbia, Marco; Fregonese, Sebastien; Deng, Marina; Curutchet, Arnaud; Yadav, Chandan; Celi, Didier; De Matos, Magali; Zimmer, Thomas

被引用次数: 1

2021

Importance of Probe Choice for Extracting Figures of Merit of Advanced mmW Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fregonese, Sebastien; De Matos, Magali; Deng, Marina; Celi, Didier; Derrier, Nicolas; Zimmer, Thomas

被引用次数: 5

2021

Extraction of True Finger Temperature From Measured Data in Multifinger Bipolar Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Gupta, Aakashdeep; Nidhin, K.; Balanethiram, Suresh; D'Esposito, Rosario; Fregonese, Sebastien; Zimmer, Thomas; Chakravorty, Anjan

被引用次数: 3

2021

Investigation of Variation in On-Si On-Wafer TRL Calibration in Sub-THz

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Yadav, Chandan; Deng, Marina; Fregonese, Sebastien; Cabbia, Marco; De Matos, Magali; Zimmer, Thomas
被引用次数: 4 2021

Sub-THz and THz SiGe HBT Electrical Compact Modeling

ELECTRONICS

Saha, Bishwadeep; Fregonese, Sebastien; Chakravorty, Anjan; Panda, Soumya Ranjan; Zimmer, Thomas
被引用次数: 1 2021

Terahertz refractive index-based morphological dilation for breast carcinoma delineation

SCIENTIFIC REPORTS

Cassar, Quentin; Caravera, Samuel; MacGrogan, Gaetan; Buecher, Thomas; Hillger, Philipp; Pfeiffer, Ullrich; Zimmer, Thomas; Guillet, Jean-Paul; Mounaix, Patrick

被引用次数: 23 2021

Guideline for Test-Structures Placement for on-Wafer Calibration in sub-THz Si Device Characterization

2021 IEEE MTT-S INTERNATIONAL MICROWAVE SYMPOSIUM (IMS)

Yadav, Chandan; Cabbia, Marco; Fregonese, Sebastien; Deng, Marina; De Matos, Magali; Zimmer, Thomas
被引用次数: 2 2021

Association of the terahertz refractive index and morphological dilation operations for breast carcinoma detection

2021 46TH INTERNATIONAL CONFERENCE ON INFRARED, MILLIMETER AND TERAHERTZ WAVES (IRMMW-THZ)

Cassar, Q.; Caravera, S.; MacGrogan, G.; Bucher, T.; Hillger, P.; Pfeiffer, U.; Zimmer, T.; Guillet, J. P.; Mounaix, P.

被引用次数: 1 2021

Electro-thermal limitations and device degradation of SiGe HBTs with emphasis on circuit performance

2021 IEEE BICMOS AND COMPOUND SEMICONDUCTOR INTEGRATED CIRCUITS AND TECHNOLOGY SYMPOSIUM (BCICTS)

Fregonese, Sebastien; Mukherjee, Chhandak; Ruecker, Holger; Chevalier, Pascal; Fischer, Gerhard; Celi, Didier; Deng, Marina; Couret, Marine; Marc, Francois; Maneux, Cristell; Zimmer, Thomas

被引用次数: 1 2021

TCAD and EM co-simulation method to verify SiGe HBT measurements up to 500 GHz

SOLID-STATE ELECTRONICS

Panda, Soumya Ranjan; Fregonese, Sebastien; Deng, Marina; Chakravorty, Anjan; Zimmer, Thomas

被引用次数: 6 2020

Scalable compact modeling of trap generation near the EB spacer oxide interface in SiGe HBTs

SOLID-STATE ELECTRONICS

Couret, Marine; Jaoul, Mathieu; Marc, Francois; Mukherjee, Chhandak; Celi, Didier; Zimmer, Thomas; Maneux, Cristell

被引用次数: 3 2020

A physical and versatile aging compact model for hot carrier degradation in SiGe HBTs under dynamic operating conditions

SOLID-STATE ELECTRONICS

Mukherjee, C.; Marc, F.; Couret, M.; Fischer, G. G.; Jaoul, M.; Celi, D.; Aufinger, K.; Zimmer, T.; Maneux, C.

被引用次数: 8

2020

High-Frequency Noise Characterization and Modeling of Graphene Field-Effect Transistors

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Deng, Marina; Fadil, Dalal; Wei, Wei; Pallecchi, Emiliano; Happy, Henri; Dambrine, Gilles; De Matos, Magali; Zimmer, Thomas; Fregonese, Sebastien

被引用次数: 7

2020

Importance and Requirement of Frequency Band Specific RF Probes EM Models in Sub-THz and THz Measurements up to 500 GHz

IEEE TRANSACTIONS ON TERAHERTZ SCIENCE AND TECHNOLOGY

Yadav, Chandan; Deng, Marina; Fregonese, Sebastien; Cabbia, Marco; De Matos, Magali; Plano, Bernard; Zimmer, Thomas

被引用次数: 22

2020

Design of On-Wafer TRL Calibration Kit for InP Technologies Characterization up to 500 GHz

IEEE TRANSACTIONS ON ELECTRON DEVICES

Deng, Marina; Mukherjee, Chhandak; Yadav, Chandan; Fregonese, Sebastien; Zimmer, Thomas; De Matos, Magali; Quan, Wei; Arabhavi, Akshay Mahadev; Bolognesi, Colombo R.; Wen, Xin; Luisier, Mathieu; Raya, Christian; Ardouin, Bertrand; Maneux, Cristell

被引用次数: 5

2020

Silicon Test Structures Design for Sub-THz and THz Measurements

IEEE TRANSACTIONS ON ELECTRON DEVICES

Cabbia, Marco; Yadav, Chandan; Deng, Marina; Fregonese, Sebastien; De Matos, Magali; Zimmer, Thomas

被引用次数: 12

2020

Analysis of High-Frequency Measurement of Transistors Along With Electromagnetic and SPICE Cosimulation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fregonese, Sebastien; Cabbia, Marco; Yadav, Chandan; Deng, Marina; Panda, Soumya Ranjan; De Matos, Magali; Celi, Didier; Chakravorty, Anjan; Zimmer, Thomas

被引用次数: 12

2020

An Efficient Thermal Model for Multifinger SiGe HBTs Under Real Operating Condition

IEEE TRANSACTIONS ON ELECTRON DEVICES

Nidhin, K.; Pande, Shubham; Yadav, Shon; Balanethiram, Suresh; Nair, Deleep R.; Fregonese, Sebastien; Zimmer, Thomas; Chakravorty, Anjan

被引用次数: 4

2020

THz Characterization and Modeling of SiGe HBTs: Review (Invited)

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Fregonese, Sebastien; Deng, Marina; Cabbia, Marco; Yadav, Chandan; De Matos, Magali; Zimmer, Thomas
被引用次数: 3 2020

Static Thermal Coupling Factors in Multi-Finger Bipolar Transistors: Part I-Model Development

ELECTRONICS

Gupta, Aakashdeep; Nidhin, K.; Balanethiram, Suresh; Yadav, Shon; Chakravorty, Anjan; Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 3 2020

Static Thermal Coupling Factors in Multi-Finger Bipolar Transistors: Part II-Experimental Validation

ELECTRONICS

Gupta, Aakashdeep; Nidhin, K.; Balanethiram, Suresh; Yadav, Shon; Chakravorty, Anjan; Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 1 2020

A Broadband Active Microwave Monolithically Integrated Circuit Balun in Graphene Technology

APPLIED SCIENCES-BASEL

Fadil, Dalal; Passi, Vikram; Wei, Wei; Ben Salk, Soukaina; Zhou, Di; Strupinski, Wlodek; Lemme, Max C.; Zimmer, Thomas; Pallecchi, Emiliano; Happy, Henri; Fregonese, Sebastien

被引用次数: 4 2020

Ex Vivo Breast Tumor Identification: Advances Toward a Silicon-Based Terahertz Near-Field Imaging Sensor

IEEE MICROWAVE MAGAZINE

Pfeiffer, Ullrich R.; Hillger, Philipp; Jain, Ritesh; Grzyb, Janusz; Bucher, Thomas; Cassar, Quentin; MacGrogan, Gaetan; Guillet, Jean-Paul; Mounaix, Patrick; Zimmer, Thomas

被引用次数: 11 2019

Comparison of On-Wafer TRL Calibration to ISS SOLT Calibration With Open-Short De-Embedding up to 500 GHz

IEEE TRANSACTIONS ON TERAHERTZ SCIENCE AND TECHNOLOGY

Fregonese, Sebastien; Deng, Marina; De Matos, Magali; Yadav, Chandan; Joly, Simon; Plano, Bernard; Raya, Christian; Ardouin, Bertrand; Zimmer, Thomas

被引用次数: 42 2019

A Multiscale TCAD Approach for the Simulation of InP DHBTs and the Extraction of Their Transit Times

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wen, Xin; Mukherjee, Chhandak; Raya, Christian; Ardouin, Bertrand; Deng, Marina; Fregonese, Sebastien; Nodjiadjim, Virginie; Riet, Muriel; Quan, Wei; Arabhavi, Akshay; Ostinelli, Olivier; Bolognesi, Colombo; Zimmer, Thomas; Maneux, Cristell; Luisier, Mathieu

被引用次数: 12 2019

Validation of Thermal Resistance Extracted From Measurements on Stripe Geometry SiGe HBTs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Balanethiram, Suresh; D'Esposito, Rosario; Fregonese, Sebastien; Chakravorty, Anjan; Zimmer, Thomas

被引用次数: 4

2019

Scalable Modeling of Thermal Impedance in InP DHBTs Targeting Terahertz Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Mukherjee, Chhandak; Couret, Marine; Nodjiadjim, Virginie; Riet, Muriel; Dupuy, J. -Y.; Fregonese, Sebastien; Zimmer, Thomas; Maneux, Cristell

被引用次数: 6

2019

A Compact Formulation for Avalanche Multiplication in SiGe HBTs at High Injection Levels

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jaoul, Mathieu; Maneux, Cristell; Celi, Didier; Schroeter, Michael; Zimmer, Thomas

被引用次数: 14

2019

A 128-Pixel System-on-a-Chip for Real-Time Super-Resolution Terahertz Near-Field Imaging

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hillger, Philipp; Jain, Ritesh; Grzyb, Janusz; Foerster, Wolfgang; Heinemann, Bernd; MacGrogan, Gaetan; Mounaix, Patrick; Zimmer, Thomas; Pfeiffer, Ullrich R.

被引用次数: 33

2018

On-Wafer Characterization of Silicon Transistors Up To 500 GHz and Analysis of Measurement Discontinuities Between the Frequency Bands

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Fregonese, Sebastien; De Matos, Magali; Deng, Marina; Potereau, Manuel; Ayela, Cedric; Aufinger, Klaus; Zimmer, Thomas

被引用次数: 33

2018

Scalable Compact Modeling of III-V DHBTs: Prospective Figures of Merit Toward Terahertz Operation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Mukherjee, Chhandak; Raya, Christian; Ardouin, Bertrand; Deng, Marina; Fregonese, Sebastien; Zimmer, Thomas; Nodjiadjim, Virginie; Riet, Muriel; Dupuy, Jean-Yves; Luisier, Mathieu; Quan, Wei; Arabhavi, Akshay; Bolognesi, Colombo R.; Maneux, Cristell

被引用次数: 16

2018

Charge-Based Modeling of Transition Metal Dichalcogenide Transistors Including Ambipolar, Trapping, and Negative Capacitance Effects

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yadav, Chandan; Rastogi, Priyank; Zimmer, Thomas; Chauhan, Yogesh Singh

被引用次数: 17

2018

Analysis of Electrothermal and Impact-Ionization Effects in Bipolar Cascode Amplifiers

IEEE TRANSACTIONS ON ELECTRON DEVICES

d'Alessandro, Vincenzo; D'Esposito, Rosario; Metzger, Andre G.; Kwok, Kai H.; Aufinger, Klaus; Zimmer, Thomas; Rinaldi, Niccolo

被引用次数: 9

2018

Pilot study of freshly excised breast tissue response in the 300-600 GHz range

BIOMEDICAL OPTICS EXPRESS

Cassar, Quentin; Al-Ibadi, Amel; Mavarani, Laven; Hillger, Philipp; Grzyb, Janusz; MacGrogan, Gaetan; Zimmer, Thomas; Pfeiffer, Ullrich R.; Guillet, Jean-Paul; Mounaix, Patrick

被引用次数: 52

2018

Enhanced Intrinsic Voltage Gain in Artificially Stacked Bilayer CVD Graphene Field Effect Transistors

ANNALEN DER PHYSIK

Pandey, Himadri; Aguirre-Morales, Jorge-Daniel; Kataria, Satender; Fregonese, Sebastien; Passi, Vikram; Iannazzo, Mario; Zimmer, Thomas; Alarcon, Eduard; Lemme, Max C.

被引用次数: 4

2017

Thermal Penetration Depth Analysis and Impact of the BEOL Metals on the Thermal Impedance of SiGe HBTs

IEEE ELECTRON DEVICE LETTERS

D'Esposito, Rosario; Balanethiram, Suresh; Battaglia, Jean-Luc; Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 17

2017

Hot-Carrier Degradation in SiGe HBTs: A Physical and Versatile Aging Compact Model

IEEE TRANSACTIONS ON ELECTRON DEVICES

Mukherjee, Chhandak; Jacquet, Thomas; Fischer, Gerhard G.; Zimmer, Thomas; Maneux, Cristell

被引用次数: 22

2017

A Large-Signal Monolayer Graphene Field-Effect Transistor Compact Model for RF-Circuit Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Aguirre-Morales, Jorge-Daniel; Fregonese, Sebastien; Mukherjee, Chhandak; Wei, Wei; Happy, Henri; Maneux, Cristell; Zimmer, Thomas

被引用次数: 21

2017

Accurate Modeling of Thermal Resistance for On-Wafer SiGe HBTs Using Average Thermal Conductivity

IEEE TRANSACTIONS ON ELECTRON DEVICES

Balanethiram, Suresh; Chakravorty, Anjan; D'Esposito, Rosario; Fregonese, Sebastien; Celi, Didier; Zimmer, Thomas

被引用次数: 15

2017

Microscopic Hot-Carrier Degradation Modeling of SiGe HBTs Under Stress Conditions Close to the SOA Limit

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kamrani, Hamed; Jabs, Dominic; d'Alessandro, Vincenzo; Rinaldi, Niccolo; Jacquet, Thomas; Maneux, Cristell; Zimmer, Thomas; Aufinger, Klaus; Jungemann, Christoph

被引用次数: 21

2017

Extraction of BEOL Contributions for Thermal Resistance in SiGe HBTs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Balanethiram, Suresh; D'Esposito, Rosario; Chakravorty, Anjan; Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 12

2017

Reliability-Aware Circuit Design Methodology for Beyond-5G Communication Systems

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Mukherjee, Chhandak; Ardouin, Bertrand; Dupuy, Jean-Yves; Nodjiadjim, Virginie; Riet, Muriel; Zimmer, Thomas; Marc, Francois; Maneux, Cristell

被引用次数: 23

2017

Random telegraph noise in SiGe HBTs: Reliability analysis close to SOA limit

MICROELECTRONICS RELIABILITY

Mukherjee, C.; Jacquet, T.; Chakravorty, A.; Zimmer, T.; Boeck, J.; Aufinger, K.; Maneux, C.

被引用次数: 4

2017

Effects of BEOL on self-heating and thermal coupling in SiGe multi-finger HBTs under real operating condition

SOLID-STATE ELECTRONICS

Dwivedi, A. D. D.; Chakravorty, Anjan; D'Esposito, Rosario; Sahoo, Amit Kumar; Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 13

2016

On the development of a novel high VSWR programmable impedance tuner

INTERNATIONAL JOURNAL OF MICROWAVE AND WIRELESS TECHNOLOGIES

Curutchet, Arnaud; Ghiotto, Anthony; Potereau, Manuel; De Matos, Magali; Fregonese, Sebastien; Kerherve, Eric; Zimmer, Thomas

被引用次数: 0

2016

Efficient Modeling of Distributed Dynamic Self-Heating and Thermal Coupling in Multifinger SiGe HBTs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Balanethiram, Suresh; D'Esposito, Rosario; Chakravorty, Anjan; Fregonese, Sebastien; Celi, Didier; Zimmer, Thomas

被引用次数: 4

2016

Analytic Estimation of Thermal Resistance in HBTs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Chakravorty, Anjan; D'Esposito, Rosario; Balanethiram, Suresh; Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 15

2016

Innovative SiGe HBT Topologies With Improved Electrothermal Behavior

IEEE TRANSACTIONS ON ELECTRON DEVICES

D'Esposito, Rosario; Fregonese, Sebastien; Chakravorty, Anjan; Chevalier, Pascal; Celi, Didier; Zimmer, Thomas

被引用次数: 8

2016

Comments on Optimization of a Compact I-V Model for Graphene FETs: Extending Parameter Scalability for Circuit Design Exploration

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 2

2016

A Study on Self-Heating and Mutual Thermal Coupling in SiGe Multi-Finger HBTs

JOURNAL OF ELECTRONIC MATERIALS

Dwivedi, A. D. D.; D'Esposito, Rosario; Sahoo, Amit Kumar; Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 3

2016

Obtaining DC and AC isothermal electrical characteristics for RF MOSFET

SOLID-STATE ELECTRONICS

Sahoo, A. K.; Fregonese, S.; Scheer, P.; Celi, D.; Juge, A.; Zimmer, T.

被引用次数: 0

2015

A Geometry Scalable Model for Nonlinear Thermal Impedance of Trench Isolated HBTs

IEEE ELECTRON DEVICE LETTERS

Sahoo, Amit Kumar; Fregonese, Sebastien; Desposito, Rosario; Aufinger, Klaus; Maneux, Cristell; Zimmer, Thomas

被引用次数: 11

2015

An Accurate Physics-Based Compact Model for Dual-Gate Bilayer Graphene FETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Aguirre-Morales, Jorge-Daniel; Fregonese, Sebastien; Mukherjee, Chhandak; Maneux, Cristell; Zimmer, Thomas

被引用次数: 18

2015

Graphene Transistor-Based Active Balun Architectures

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zimmer, Thomas; Fregonese, Sebastien

被引用次数: 11

2015

Versatile Compact Model for Graphene FET Targeting Reliability-Aware Circuit Design

IEEE TRANSACTIONS ON ELECTRON DEVICES

Mukherjee, Chhandak; Aguirre-Morales, Jorge-Daniel; Fregonese, Sebastien; Zimmer, Thomas; Maneux, Cristell

被引用次数: 19

2015

Isothermal Electrical Characteristic Extraction for mmWave HBTs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sahoo, Amit Kumar; Fregonese, Sebastien; D'Esposito, Rosario; Maneux, Cristell; Zimmer, Thomas

被引用次数: 1

2015

Source-Pull and Load-Pull Characterization of Graphene FET

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Fregonese, Sebastien; De Matos, Magali; Mele, David; Maneux, Cristell; Happy, Henri; Zimmer, Thomas
被引用次数: 6 2015

Electrical Compact Modeling of Graphene Base Transistors

ELECTRONICS

Fregonese, Sebastien; Venica, Stefano; Driussi, Francesco; Zimmer, Thomas

被引用次数: 9 2015

Innovative Dual-Gate CNTFET Logic Cell: Investigation of Technological Dispersion Impact Through Compact Modeling

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Maneux, Cristell; Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 6 2014

Multiscale simulation of carbon nanotube transistors

SOLID-STATE ELECTRONICS

Maneux, Cristell; Fregonese, Sebastien; Zimmer, Thomas; Retailleau, Sylvie; Huu Nha Nguyen; Querlioz, Damien; Bournel, Arnaud; Dollfus, Philippe; Triozon, Francois; Niquet, Yann Michel; Roche, Stephan

被引用次数: 16 2013

80 ns/45 GHz Pulsed measurement system for DC and RF characterization of high speed microwave devices

SOLID-STATE ELECTRONICS

Weiss, Mario; Fregonese, Sebastien; Santorelli, Marco; Sahoo, Amit Kumar; Maneux, Cristell; Zimmer, Thomas

被引用次数: 6 2013

Foreword

SOLID-STATE ELECTRONICS

Zimmer, Thomas; Fregonese, Sebastien

被引用次数: 0 2013

Optimized Ring Oscillator With 1.65-ps Gate Delay in a SiGe:C HBT Technology

IEEE ELECTRON DEVICE LETTERS

Weiss, Mario; Majek, Cedric; Sahoo, Amit Kumar; Maneux, Cristell; Mazouffre, Olivier; Chevalier, Pascal; Chantre, Alain; Zimmer, Thomas

被引用次数: 6 2013

Robust Surface-Potential-Based Compact Model for GaN HEMT IC Design

IEEE TRANSACTIONS ON ELECTRON DEVICES

Khandelwal, Sourabh; Yadav, Chandan; Agnihotri, Shantanu; Chauhan, Yogesh Singh; Curutchet, Arnaud; Zimmer, Thomas; De Jaeger, Jean-Claude; Defrance, Nicolas; Fjeldly, Tor A.

被引用次数: 95 2013

Submicrometer InP/InGaAs DHBT Architecture Enhancements Targeting Reliability Improvements

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kone, Gilles A.; Grandchamp, Brice; Hainaut, Cyril; Marc, Francois; Labat, Nathalie; Zimmer, Thomas; Nodjadjim, Virginie; Riet, Muriel; Dupuy, Jean-Yves; Godin, Jean; Maneux, Cristell

被引用次数: 10

2013

Scalable Electrical Compact Modeling for Graphene FET Transistors

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Fregonese, Sebastien; Magallo, Maura; Maneux, Cristell; Happy, Henri; Zimmer, Thomas

被引用次数: 97

2013

Benchmarking of GFET devices for amplifier application using multiscale simulation approach

JOURNAL OF COMPUTATIONAL ELECTRONICS

Fregonese, Sebastien; Potereau, Manuel; Deltimple, Nathalie; Maneux, Cristell; Zimmer, Thomas

被引用次数: 7

2013

Transient electro-thermal characterization of Si-Ge heterojunction bipolar transistors

SOLID-STATE ELECTRONICS

Sahoo, Amit Kumar; Weiss, Mario; Fregonese, Sebastien; Malbert, Nathalie; Zimmer, Thomas

被引用次数: 8

2012

Electrical compact modelling of graphene transistors

SOLID-STATE ELECTRONICS

Fregonese, Sebastien; Meng, Nan; Huu-Nha Nguyen; Majek, Cedric; Maneux, Cristell; Happy, Henri; Zimmer, Thomas

被引用次数: 31

2012

A Scalable Electrothermal Model for Transient Self-Heating Effects in Trench-Isolated SiGe HBTs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sahoo, Amit Kumar; Fregonese, Sebastien; Weiss, Mario; Malbert, Nathalie; Zimmer, Thomas

被引用次数: 30

2012

Modeling Non-Quasi-Static Effects in SiGe HBTs Using Improved Charge Partitioning Scheme

IEEE TRANSACTIONS ON ELECTRON DEVICES

Augustine, Noel; Kumar, Khamesh; Bhattacharyya, Arkaprava; Zimmer, Thomas; Chakravorty, Anjan

被引用次数: 3

2012

Characterization and Modeling of Graphene Transistor Low-Frequency Noise

IEEE TRANSACTIONS ON ELECTRON DEVICES

Grandchamp, B.; Fregonese, S.; Majek, C.; Hainaut, C.; Maneux, C.; Meng, N.; Happy, H.; Zimmer, T.

被引用次数: 17

2012

Design and Modeling of a Neuro-Inspired Learning Circuit Using Nanotube-Based Memory Devices

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Liao, Si-Yu; Retrouvey, Jean-Marie; Agnus, Guillaume; Zhao, Weisheng; Maneux, Cristell; Fregonese, Sebastien; Zimmer, Thomas; Chabi, Djaafar; Filoramo, Arianna; Derycke, Vincent; Gamrat, Christian; Klein, Jacques-Olivier

被引用次数: 12

2011

Trends in Submicrometer InP-Based HBT Architecture Targeting Thermal Management

IEEE TRANSACTIONS ON ELECTRON DEVICES

Grandchamp, Brice; Nodjiadjim, Virginie; Zaknoune, Mohammed; Kone, Gilles A.; Hainaut, Cyril; Godin, Jean; Riet, Muriel; Zimmer, Thomas; Maneux, Cristell

被引用次数: 14

2011

A Compact Model for Dual-Gate One-Dimensional FET: Application to Carbon-Nanotube FETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fregonese, Sebastien; Maneux, Cristell; Zimmer, Thomas

被引用次数: 27

2011

Schottky Barrier Carbon Nanotube Transistor: Compact Modeling, Scaling Study, and Circuit Design Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Najari, Montassar; Fregonese, Sebastien; Maneux, Cristell; Mnif, Hassene; Masmoudi, Nouri; Zimmer, Thomas

被引用次数: 42

2011

Reliability of submicron InGaAs/InP DHBT under thermal and electrical stresses

MICROELECTRONICS RELIABILITY

Kone, G. A.; Grandchamp, B.; Hainaut, C.; Marc, F.; Maneux, C.; Labat, N.; Zimmer, T.; Nodjiadjim, V.; Riet, M.; Godin, J.

被引用次数: 7

2011

Investigation of the degradation mechanisms of InP/InGaAs DHBT under bias stress conditions to achieve electrical aging model for circuit design

MICROELECTRONICS RELIABILITY

Ghosh, S.; Grandchamp, B.; Kone, G. A.; Marc, F.; Maneux, C.; Zimmer, T.; Nodjiadjim, V.; Riet, M.; Dupuy, J. - Y.; Godin, J.

被引用次数: 8

2011

A versatile compact model for ballistic 1D transistor: GNR-FET and CNT-FET comparison

SOLID-STATE ELECTRONICS

Fregonese, Sebastien; Maneux, Cristell; Zimmer, Thomas

被引用次数: 7

2010

Compact modeling of optically gated carbon nanotube field effect transistor

PHYSICA STATUS SOLIDI B-BASIC SOLID STATE PHYSICS

Liao, Si-Yu; Maneux, Cristell; Pouget, Vincent; Fregonese, Sebastien; Zimmer, Thomas

被引用次数: 4

2010

TCAD modeling of NPN-SI-BJT electrical performance improvement through SiGe extrinsic stress layer

MATERIALS SCIENCE IN SEMICONDUCTOR PROCESSING

Al-Sa'di, Mahmoud; Fregonese, Sebastien; Maneux, Cristell; Zimmer, Thomas

被引用次数: 4

2010

Preliminary results of storage accelerated aging test on InP/InGaAs DHBT

MICROELECTRONICS RELIABILITY

Kone, G. A.; Grandchamp, B.; Hainaut, C.; Marc, F.; Maneux, C.; Labat, N.; Zimmer, T.; Nodjiadjim, V.; Godin, J.

被引用次数: 7

2010

Thermal aging model of InP/InGaAs/InP DHBT

MICROELECTRONICS RELIABILITY

Ghosh, S.; Marc, F.; Maneux, C.; Grandchamp, B.; Kone, G. A.; Zimmer, T.

被引用次数: 5

2010

Technological dispersion in CNTFET: Impact of the presence of metallic carbon nanotubes in logic circuits

SOLID-STATE ELECTRONICS

Fregonese, Sebastien; Maneux, Cristell; Zimmer, Thomas

被引用次数: 3

2009

Multiscale simulation of carbon nanotube devices

COMPTES RENDUS PHYSIQUE

Adessi, C.; Avriller, R.; Blase, X.; Bournel, A.; d'Honincthun, H. Cazin; Dollfus, P.; Fregonese, S.; Galdin-Retailleau, S.; Lopez-Bezanilla, A.; Maneux, C.; Nguyen, H. Nha; Querlioz, D.; Roche, S.; Triozon, F.; Zimmer, T.

被引用次数: 12

2009

Implementation of Tunneling Phenomena in a CNTFET Compact Model

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fregonese, Sebastien; Maneux, Cristell; Zimmer, Thomas

被引用次数: 22

2009

Implementation of Electron-Phonon Scattering in a CNTFET Compact Model

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fregonese, Sebastien; Goguet, Johnny; Maneux, Cristell; Zimmer, Thomas

被引用次数: 24

2009

A Distance Measurement Platform Dedicated to Electrical Engineering

IEEE TRANSACTIONS ON LEARNING TECHNOLOGIES

Lewis, Noelle; Billaud, Michel; Geoffroy, Didier; Cazenave, Philippe; Zimmer, Thomas

被引用次数: 16

2009

Challenges and potential of new approaches for reliability assessment of nanotechnologies

COMPTES RENDUS PHYSIQUE

Bechou, Laurent; Danto, Yves; Deletage, Jean-Yves; Verdier, Frederic; Deshayes, Yannick; Fregonese, Sbastien; Maneux, Cristell; Zimmer, Thomas; Laffitte, Dominique

被引用次数: 4

2008

Computationally efficient physics-based compact CNTFET model for circuit design

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fregonese, Sebastien; d'Honincthun, Hugues Cazin; Goguet, Johnny; Maneux, Cristell; Zimmer, Thomas; Bourgoin, Jean-Philippe; Dollfus, Philippe; Galdin-Retailleau, Sylvie

被引用次数: 73

2008

Electrical behavior and technology optimization of Si/SiGeC HBTs on thin-film SOI

IEEE TRANSACTIONS ON ELECTRON DEVICES

Avenier, Gregory; Fregonese, Sebastien; Chevalier, Pascal; Bustos, Jessy; Saguin, Fabienne; Schwartzmann, Thierry; Maneux, Cristell; Zimmer, Thomas; Chantre, Alain

被引用次数: 24

2008

A nodal model dedicated to self-heating and thermal coupling simulations

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Beckrich-Ros, Helene; Ortolland, Sylvie; Pache, Denis; Celi, Didier; Gloria, Daniel; Zimmer, Thomas

被引用次数: 5

2008

Scalable approach for HBT's base resistance calculation

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Raya, Christian; Pourchon, Franck; Zimmer, Thomas; Celi, Didier; Chevalier, Pascal

被引用次数: 13

2008

CNTFET modeling and reconfigurable logic-circuit design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

O'Connor, Ian; Liu, Junchen; Pregaldiny, Fabien; Lallement, Christophe; Maneux, Cristell; Goguet, Johnny; Fregonese, Sebastien; Zimmer, Thomas; Anghel, Lorena; Dang, Trong-Trinh; Leveugle, Regis

被引用次数: 122

2007

Self heating modeling of SiGe heterojunction bipolar transistor

INTERNATIONAL COMMUNICATIONS IN HEAT AND MASS TRANSFER

Sulima, Pierre Yvan; Battaglia, Jean-Luc; Zimmer, Thomas; Batsale, J.-C.

被引用次数: 7

2007

A compact model for SiGe HBT on thin-film SOI

IEEE TRANSACTIONS ON ELECTRON DEVICES

Frégonèse, S; Avenier, G; Maneux, C; Chantre, A; Zimmer, T

被引用次数: 18

2006

A scalable substrate network for compact modelling of deep trench insulated HBT

SOLID-STATE ELECTRONICS

Fregonese, S; Celi, D; Zimmer, T; Maneux, C; Sulima, PY

被引用次数: 4

2005

An analog circuit fault characterization methodology

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Maidon, Y; Zimmer, T; Ivanov, A

被引用次数: 2

2005

Obtaining isothermal data for HBT

IEEE TRANSACTIONS ON ELECTRON DEVICES

Frégonèse, S; Zimmer, T; Mnif, H; Baureis, P; Maneux, C

被引用次数: 2

2004

Representation of the SiGe HBT's thermal impedance by linear and recursive networks

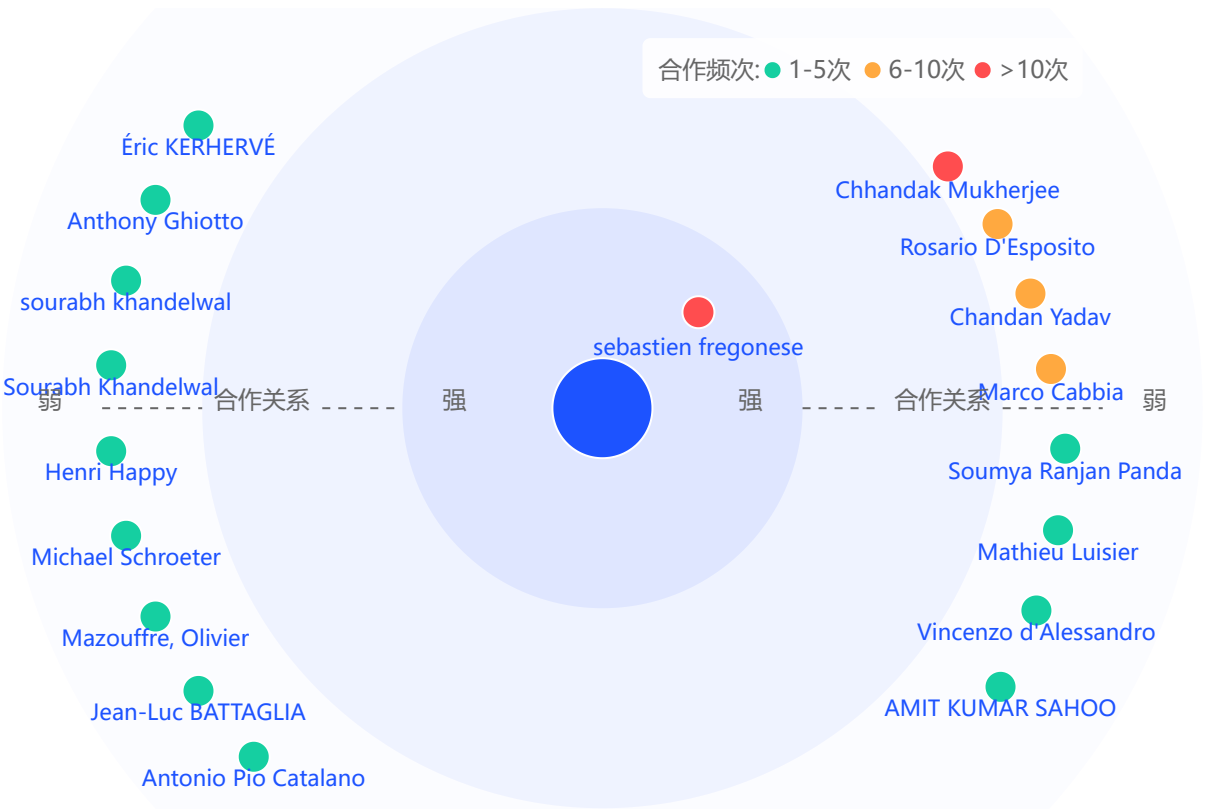
MICROELECTRONICS RELIABILITY

Mnif, H; Zimmer, T; Battaglia, JL; Fregonese, S

被引用次数: 13

2004

学术合作



sebastien fregonese	合作次数: 51
波尔多大学	
Chhandak Mukherjee	合作次数: 11

波尔多大学

Rosario D'Esposito

合作次数: 9

波尔多大学

Chandan Yadav

合作次数: 8

波尔多大学

Marco Cabbia

合作次数: 7

波尔多大学

还有 13 个合作者